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Innate immune defenses at the maternal-fetal interface

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Abstract

The human maternal-fetal interface is an immunologically complex environment that must balance the divergent demands of tolerance towards the developing fetus with anti-pathogen defense. The innate immune responses at the maternal-fetal interface that function in anti-microbial defense have been understudied to-date and how “TORCH” pathogens evade maternal innate immunity to infect the fetus remains poorly understood. Herein, we discuss how newly described decidual innate lymphoid cells and maternal placenta-associated macrophage subsets may be involved in anti-pathogen defense. Moreover, we outline recent advances in our understanding of how placental trophoblasts and fetal-derived macrophages (Hofbauer cells) function in anti-microbial defense. In summary, we highlight current gaps in knowledge and describe novel experimental models of the human decidua and placenta that are poised to advance our knowledge of innate immune defenses at the maternal-fetal interface.

Keywords

placenta; immunity; Maternal-fetal interface; TORCH; Decidua

Introduction

The maternal-fetal interface is an immunologically dynamic environment composed of the maternal-derived decidua and fetal-derived placenta (Figure 1; reviewed in [1]). Throughout gestation, the maternal-fetal interface must balance the divergent demands of immunotolerance towards the semi-allogenic fetus and immune defense against infections. While the maternal host and placenta are able to successfully prevent most pathogens from infecting the fetus, select pathogens collectively referred to as “TORCH” (including

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Toxoplasma gondii, Other (Zika virus (ZIKV), human immunodeficiency virus (HIV)), Rubella virus, human Cytomegalovirus (HCMV), and Herpesviruses) are able to evade maternal-fetal immune responses to cross the placental barrier and infect the fetus (reviewed in [2,3]). In this concise review, we discuss recent advances regarding decidual and placental innate immune cells and innate immune defenses employed by the maternal-fetal interface to defend against infections. We focus on antimicrobial innate immune responses rather than the immune signaling involved in healthy maternal-fetal crosstalk and emphasize findings from human studies given differences between human and murine placental biology (reviewed in [4]). Finally, we discuss recent advances in human models of the maternal-fetal interface that are poised to transform the field.

Decidual innate immunity

The role of decidual innate immune responses in antimicrobial defense is relatively understudied to-date; however, recent advances in single-cell sequencing have revealed that the decidua contains diverse immune and non-immune cell populations that may function in anti-pathogen defense ([5,6]; reviewed in [7]). The most abundant immune cells in the maternal decidua are decidual innate lymphoid cells (dILCs), which are mostly comprised of decidual natural killer (dNK) cells (Figure 1). Maternal macrophages are the next most abundant decidual immune cell type ([5,8]), which we discuss below in relation to other placenta-associated macrophages. Decidual dendritic cells (dDCs) are also present in low numbers ([5,9]), yet it remains unclear whether these antigen-presenting dDCs function in anti-pathogen defense at the human maternal-fetal interface.

Decidual innate lymphoid cells (dILCs)

Non-NK dILCs have recently been identified in the maternal decidua, but their functionality remains underexplored ([10]). Several groups have described a diverse range of dILC subsets using high dimensional single-cell sequencing, multi-parametric flow cytometry, and mass cytometry ([5,11,12]). These studies reveal that dILCs are phenotypically heterogeneous, present throughout gestation, difficult to ascribe to ILC type 1, 2, or 3 classifications, and challenging to define using traditional 2D flow cytometry ([5,11,12]). Some dILC subsets secrete IFN γ ([10,11]), which could prime anti-viral immune responses; however, whether dILCs respond to viral infections *in vivo* has not been explored. Additional dILC subsets have been identified that produce pro-inflammatory cytokines such as IL-22, IL-8, IL-17a, and TNF- α , which could protect against infections, yet these dILCs comprise only a very small minority of dILCs overall and seem to have low cytotoxic potential ([5,10–12]).

Recently, Vento-Tormo et al. identified three dNK cell subsets present in the first trimester decidua, which are primarily anti-inflammatory and interact with extravillous trophoblast (EVTs) to facilitate placentation and promote tolerance ([5]). EVT anchor the placenta to the decidua and are critical for remodeling the maternal microvasculature as well as immune cross-talk with dNKs. These three dNK cell subsets likely persist throughout gestation as shown by complimentary findings in term decidua ([11]). Some of these dNK cell subsets express high levels of antimicrobial peptides (*e.g.*, perforin, granzyme B, and granulysin), but how dNK cells defend against pathogens *in utero* is unclear ([5,12]).

While some *in vitro* studies suggest that dNKs can act as cytotoxic effectors against decidual HCMV infection ([13]), others demonstrate that dNKs are unable to overcome tolerogenic mechanisms to degranulate in response to HCMV-infected EVTs ([14]). Yet, one intriguing novel mechanism described recently is the ability of dNKs to transfer granulysin via nanotubules to EVTs, macrophages, and dDCs to kill *Listeria monocytogenes* ([15]) (Figure 2). This unique anti-microbial defense strategy shows how dNKs can defend against intracellular pathogens while still maintaining tolerance. Outstanding questions remain about the anti-microbial mechanisms of dNKs including whether any dNK subsets use Fc receptor mediated effector functions (*e.g.*, antibody-dependent cellular cytotoxicity) and how these unique innate immune cells balance tolerance and anti-microbial defense at the maternal-fetal interface.

Syncytiotrophoblast (STB) layer and trophoblasts

The placenta contains three fetal-derived trophoblast populations, including cytotrophoblasts (CTB), which line the placental villi during early gestation and give rise to the syncytiotrophoblast (STB) layer, and EVTs, which invade the maternal decidua to facilitate placentation and cross-talk with dNKs to promote maternal-fetal tolerance (Figure 1; reviewed in [4]). While EVTs are important for maintaining tolerance, the role of EVTs in anti-microbial defense is unclear. The placenta contains three EVT populations throughout pregnancy, which are biased towards immunotolerance and minimally responsive to IFN γ , therefore additional investigation of whether they can present antigen or function in anti-microbial defense is warranted (Figure 2; [16]).

The dense multinucleated STB layer lacks cellular junctions and has constitutive anti-microbial innate immune functions, making it a formidable immunological and physical barrier against potential vertically-transmitted pathogens ([17]). While the STB is susceptible to some pathogens, such as rift valley fever virus ([18,19]), it is generally resistant to productive viral, protozoal, and bacterial infections ([17,20,21]). Constitutively, the STB secretes antimicrobial peptides, vesicles with anti-microbial microRNAs ([20,22]), and anti-viral IFN- λ to protect against TORCH pathogens (Figure 3; [23–25]). The STB also facilitates nutrient exchange and transplacental transfer of maternal IgG, inflammatory mediators, and immune complexes (reviewed in [26]). However, some vertically-transmitted viruses may hijack this normal pathway of IgG transfer to bypass the STB through immune complex binding to the neonatal Fc receptor (FcRn) or to other placentally-expressed Fc γ receptors ([27], reviewed in [2]).

All trophoblast subtypes express pattern-recognition receptors (PRRs) including toll-like receptors (TLRs) and nod-like receptors (NLRs) that allow them to recognize and respond to placental pathogens. TLR activation leads to the release of pro-inflammatory cytokines such as IFN α , IFN β , TNF- α , IL-6, and IL-8, which can further prime innate immune responses. Recently, NLRP3 inflammasome activation and subsequent IL-1 β and IL-18 release has also emerged as key trophoblast defense strategy against *Listeria monocytogenes* and *Plasmodium falciparum* ([21,28]).

However, these trophoblast anti-microbial defense strategies might come at a significant cost to the developing fetus (reviewed in [29,30]). Two recent studies demonstrated that

immune-related IFN-induced transmembrane proteins (IFITM1, -2 and -3), which are activated by type I IFNs in the setting of placental viral infection, inhibit syncytin-mediated fusion of the STB ([31,32]). Thus, improper development of the STB due to trophoblast IFN β expression could contribute to poor placental function and fetal pathology in the setting of some TORCH infections ([31,32]). Moreover, NLRP3 inflammasome activation and increased IL-1 β expression has also been associated with increased placental and fetal pathology ([28]).

Placenta-associated maternal macrophages and monocytes (PAMMs)

Macrophages, innate tissue-resident myeloid immune cells, have traditionally been defined as M1/pro-inflammatory or M2/anti-inflammatory and can function in anti-microbial defense through direct phagocytosis, Fc receptor antibody-mediated phagocytosis, cytokine release, and innate immune cell signaling. Decidual macrophages are important for tissue remodeling and induction of labor during pregnancy (reviewed in [33]), yet their role in defense against pathogens has been poorly characterized. Increasing evidence suggests that placenta-associated maternal macrophages and monocytes (PAMMs), as coined by Thomas et al., are highly heterogeneous and have both M2/anti-inflammatory and M1/pro-inflammatory functions at the maternal-fetal interface (Figures 2–3; [5,8,34]).

Decidual and placental-villi PAMMs (differentiated by HLA-DR, FOLR2, CD9, and CCR2 expression) express a variety of TLRs ([35]), suggesting that they can respond to placental pathogens. Intriguingly, decidual PAMMs release both pro- (TNF- α , IL-1 β , IL-6, IL-8) and anti- (IL-10) inflammatory cytokines following TLR stimulation, suggesting that decidual PAMMs are not polarized towards only M1 or M2 responses ([35]). Recently, Jiang et al. identified three decidual PAMM subsets (distinguished by CCR2 and CD11c expression) that further support this interpretation that decidual PAMMs have both anti- and pro-inflammatory functions ([8]). One subset distributed throughout the decidua highly expressed MHCII, suggesting an important role in antigen-presentation. Another PAMM subset localized near EVTs appears primed for pathogen defense based on high phagocytic activity and anti-microbial gene expression ([8]). Moreover, this EVT-associated PAMM had increased expression of Fc receptors CD16/Fc γ RIII and CD32/Fc γ RII, which could boost phagocytic activity and help mediate Fc effector functions such as antibody-dependent cellular phagocytosis to defend against pathogens; however, this potential function of decidual PAMMs has not been explored.

In addition to decidual PAMMs, Thomas et al. identified two PAMMs located outside the decidua, encompassing a monocyte population in the maternal intervillous space that differentiates into placental-villi associated macrophages ([34], reviewed in [36]). These placental-villi adjacent PAMMs secrete IL-1 β and IL-6 constitutively and increase IL-1 β , IL-6 and TNF- α following TLR stimulation *ex vivo*, suggesting that they can also defend against *in utero* infections ([34]). Moreover, these PAMMs can phagocytose and kill bacteria *ex vivo* ([34]). More work is needed to understand these newly described PAMMs in the intervillous space, as they are directly exposed to bloodborne pathogens and maternal antibodies, and thus may be a key first line of defense against placentally-transmitted infections (Figures 2–3).

Fetal placental macrophages (Hofbauer cells; HBCs)

Hofbauer cells (HBCs) are fetal-derived placental macrophages located in placental villi stroma throughout gestation and are the only fetal immune cell in the placenta (reviewed in [37]). Traditionally, HBCs have been classified as M2/anti-inflammatory and considered important for placental homeostasis rather than anti-microbial defense. However, recent studies have illuminated the pleotropic nature of HBCs, demonstrating that the traditional M2 classification of HBCs is oversimplistic and that HBCs play a key role in infections at the maternal-fetal interface (reviewed in [38]). An important caveat to the limited work on HBCs to-date is that previous *ex vivo* phenotypic and functional studies of HBCs were likely contaminated by the presence of PAMMs in placental macrophage preparations, as 20–40% of isolated placental macrophages are maternal rather than fetal in origin ([34,39]). Future work excluding PAMMs from HBC isolations and leveraging novel flow cytometry gating strategies (*e.g.*, HLA allotyping and fetal-specific macrophage markers) will be key to improving our understanding of HBCs ([34], reviewed in [36]).

HBCs primarily express M2-associated markers (CD206, DC-SIGN, CD163) as well as some M1-associated activation markers (CD80, CD86, HLA-DR), though immunophenotypes vary across gestation ([34,40–42]). Despite this skewing towards an M2 phenotype, HBCs secrete M1-associated pro-inflammatory cytokines (TNF- α , IL-1 β , IL-6, IL-8) following TLR stimulation *ex vivo*, suggesting that HBCs can respond to pathogens *in vivo* ([34,40–43]). Moreover, HBCs infected with HCMV or bacteria *ex vivo* upregulate M1-associated activation markers, highlighting the plasticity of HBCs in response to infection ([39,44]). HBC release of IL-1 β following inflammasome activation may also prime other non-immune placental cells to mount an innate immune response ([45]). Intriguingly, HBCs secrete anti-inflammatory cytokines (*e.g.*, IL-10) in conjunction with pro-inflammatory cytokines, which may be an evolved mechanism to preserve maternal-fetal tolerance in the setting of infection ([34,42,43,46]).

HBCs are equipped with additional anti-microbial defenses beyond pro-inflammatory cytokine release and innate immune signaling (Figure 3). HBCs can phagocytose pathogens and employ microbiocidal mechanisms (*e.g.*, acidified lysosomes, reactive oxygen species, and cathepsin B) to kill bacteria *ex vivo* ([34,39]). Additionally, HBCs can release macrophage extracellular traps (METs) composed of DNA and anti-microbial peptides, to kill bacteria *in vitro* ([47]). Yet, the *in vivo* relevance of these HBC anti-microbial defense mechanisms remains unclear.

The role of HBCs in congenital infections has been most well-described in HCMV and ZIKV infections. While HCMV and ZIKV infection of HBCs promotes anti-viral signaling pathways (*e.g.*, RIG-I like receptor/MDA-5 signaling and interferon stimulated genes) as well as pro-inflammatory cytokine release ([44,46]), these mechanisms are not sufficient to prevent placental transmission. In fact, HBCs are susceptible to productive HCMV, ZIKV, and HIV viral replication ([44,46,48]), suggesting that these placental macrophages inadvertently function as a reservoir for vertically-transmitted viruses. How these viruses subvert HBC anti-viral immune responses deserves further study and may be key to developing interventions to prevent congenital transmission.

Another understudied area is how HBC interactions with virus-IgG immune complexes may influence anti-microbial defense. HBCs express multiple Fc γ receptors including CD64/Fc γ R1, CD32/Fc γ R2, and CD16/Fc γ R3 that bind to immune complexes; however, it is unknown if HBCs effectively employ anti-microbial Fc effector functions (e.g., antibody-dependent cellular phagocytosis) and recent evidence suggests that HBCs may have decreased sensitivity to immune complex binding ([41]). Alternatively, Fc γ R immune complex binding may promote antibody-dependent enhancement of HBC infection, as has been observed *in vitro* for ZIKV or downregulate antiviral responses through inhibitory Fc γ R2b binding, which may allow TORCH pathogens to avoid HBC innate immune responses ([27]). Intriguing outstanding questions remain about how HBCs defend against versus promote placental transmission of infections at the maternal-fetal interface (reviewed in [29]) and new strategies for isolating HBCs for downstream functional experiments will likely shed new light on this area.

Concluding remarks

Though our understanding of innate immune responses at the maternal-fetal interface is rapidly evolving, our knowledge of how decidual and placental cells defend against infections and how certain pathogens overcome these defenses remains very limited. Moreover, consistent nomenclature, standardized flow cytometry markers, and gating strategies are urgently needed for these emerging decidual and placental innate immune cell subsets. Excitingly, single-cell technologies and novel *ex vivo/in vitro* models (Box 1) of the decidual-placental interface are poised to transform our understanding of this important field of human immunobiology.

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Annotated References

* special interest

** outstanding interest

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represents an exceptional technological and conceptual advance for understanding the phenotype and function of macrophages at the maternal-fetal interface.

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Box 1.**Novel models of the human maternal-fetal interface.**

Recent advances in models of the human placenta and of the maternal-fetal interface more broadly have greatly expanded our ability to perform studies in physiologically relevant systems. The use of stem cells to generate organoids of human placental trophoblasts has advanced the field substantially, particularly given that these organoids differentiate to contain the major trophoblast subtypes present in the human placenta [49–51]. Parallel models developed from the maternal decidua thus allows for generation of organoids that model the maternal component of the interface, a major technical advance that allows for generation of matched maternal and fetal organoids from the same placental tissue [52]. Studies defining the responses of organoids that models the human maternal-fetal interface to pathogen infection will undoubtedly expand our knowledge of the unique innate immune defenses at this critical site [53]. Future studies to expand these models to include immune cell subtypes are critical to further advance our understanding of immune defenses at the maternal-fetal interface.

Highlights

- Decidual innate lymphoid cells (ILCs) such as decidual natural killer (dNKs) cells can function in anti-microbial defense by transferring anti-microbial peptides to placental trophoblasts
- Newly discovered maternal placenta-associated macrophages and monocytes (PAMMs) can phagocytose pathogens and respond to toll-like receptor (TLR) stimulation
- Anti-viral interferon secretion by placental trophoblasts inadvertently contributes to placental dysfunction in congenital infections
- Fetal-derived macrophages (Hofbauer cells) can respond to toll-like receptor (TLR) stimulation and secrete pro-inflammatory cytokines, yet may also function as viral reservoirs
- Novel organoid models of the human maternal-fetal interface and single-cell technologies should be employed to further understand decidual and placental innate immunity and how certain pathogens overcome these innate immune defenses

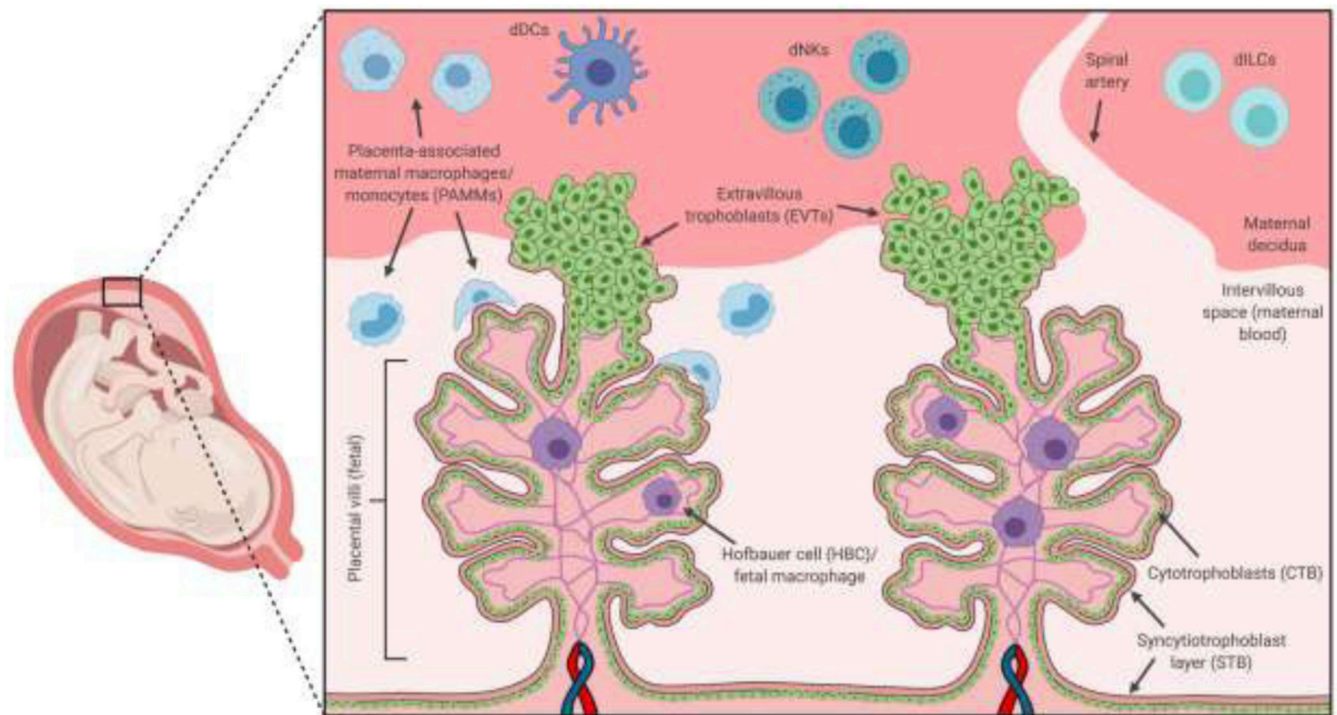


Figure 1. Overview of the maternal-fetal interface.

The maternal-fetal interface is comprised of the maternal-derived decidua and fetal-derived placental villi. Maternal-derived innate immune cells include decidual natural killer (dNK) cells, non-NK decidual innate lymphoid (dILCs), placenta-associated maternal macrophages and monocytes (PAMMs), and decidual dendritic cells (dDCs). In early gestation, the fetal placental villi become bathed in maternal blood that is delivered by the spiral arteries to form the intervillous space, which contains maternal immune cells, nutrients, antigens, antibodies, and bloodborne pathogens. The placenta contains different trophoblast populations with diverse innate immune functions including invading extravillous trophoblasts (EVTs), which remodel the spiral arteries, anchor the placental villi to the maternal decidua, and promote tolerance, as well as, cytotrophoblasts (CTB). While CTB are present in a contiguous layer surrounding the placental-villi in early gestation, these trophoblasts fuse to form the multi-nucleated syncytiotrophoblast (STB) layer later in gestation. The only fetal-derived immune cells in the placenta are primitive macrophages termed Hofbauer cells (HBCs), which are abundant throughout the placental villi from early gestation and throughout pregnancy. Figure was generated using Biorender.

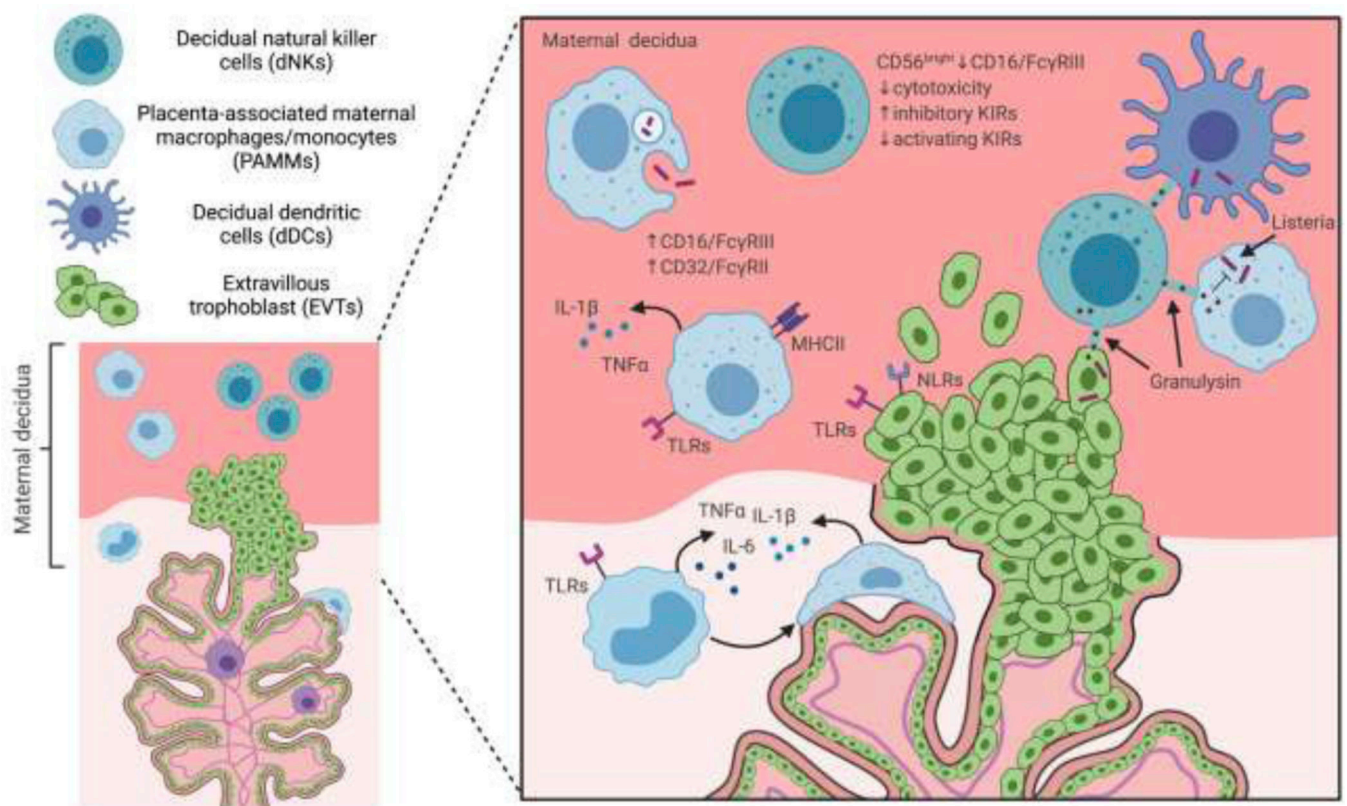


Figure 2. Decidual innate immune defenses.

Though innate immune cells in the maternal decidua are primarily anti-inflammatory and immunotolerant, there is emerging evidence that dNK and PAMM cell subsets are highly heterogeneous and may play a role in anti-microbial innate immune defense. Although dNK cells have low cytotoxic potential, they also contain granules of antimicrobial peptides (*e.g.*, perforin, granzyme, and granulysin) and can use nanotubules to transfer granulysin to EVTs to kill intracellular pathogens. Whether other anti-microbial mechanisms (*e.g.*, Fc receptor effector functions) are employed by dNKs remains unclear and should be explored. Whether EVTs function independently in anti-microbial responses is also unknown, though they express pathogen recognition receptors (PRRs) including toll-like receptors (TLRs) and nod-like receptors (NLRs) that can facilitate pathogen detection. PAMM subsets are diverse and while some are biased towards an anti-inflammatory/M2 phenotype that has classically been ascribed to maternal decidual macrophages, others have more clear pro-inflammatory/M1-like functions. For instance, PAMMs can release pro-inflammatory cytokines following TLR stimulation and present antigen through MHC class II receptors; however, whether decidual PAMMs phagocytose pathogens directly or mediate Fc receptor effector functions such as antibody-dependent cellular phagocytosis is less clear and should be investigated. Figure was generated using Biorender.

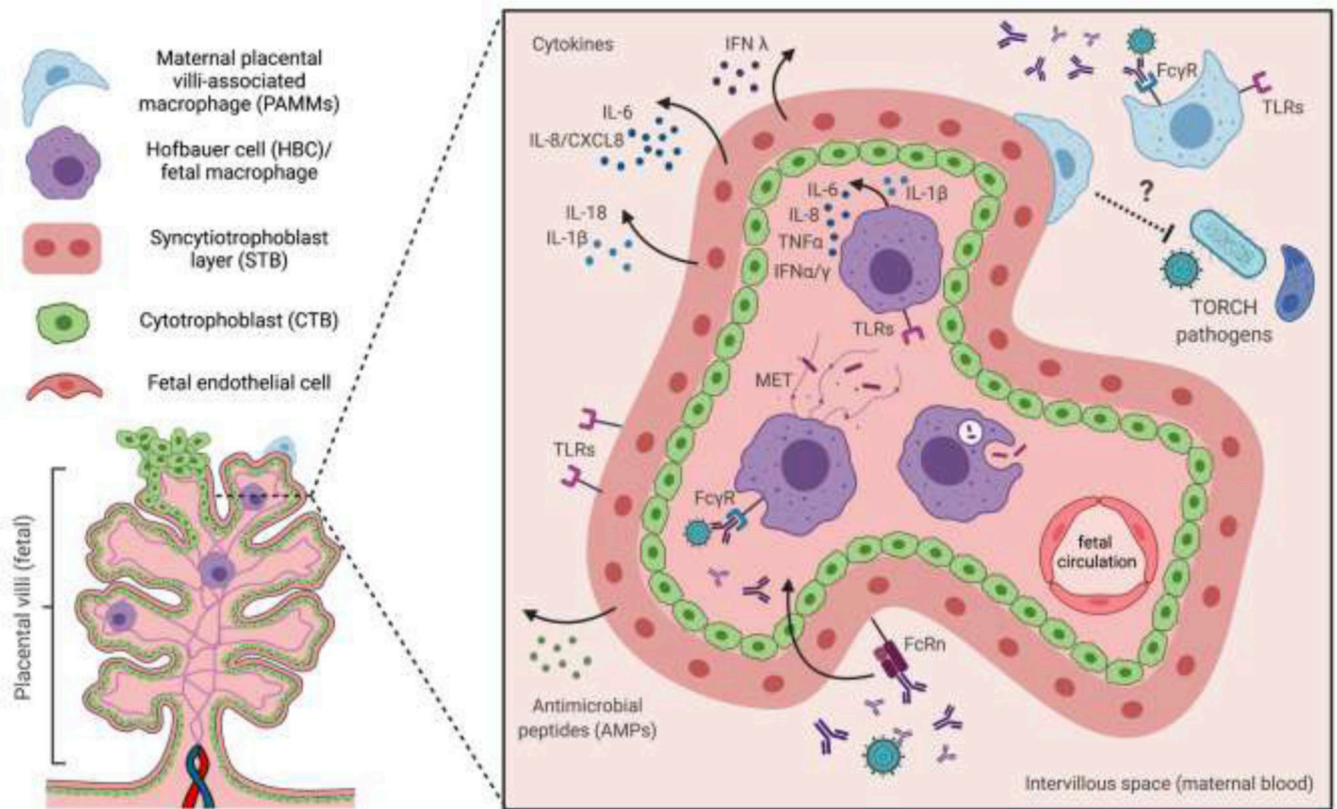


Figure 3. Placental-villi innate immune defenses.

Placental villi have robust anti-microbial defense mechanisms to prevent congenital infections; however, certain TORCH pathogens are able to subvert these innate immune defenses to infect the developing fetus. The multi-nucleated STB layer is a key physical defense against pathogens that also constitutively secretes type III interferons, microRNAs, and antimicrobial peptides. Moreover, the STB contains numerous TLRs and NLRs for pathogen detection that initiate rapid innate immune signaling. The STB also facilitates passive transplacental antibody transfer, though this may inadvertently also facilitate immune complex transfer, allowing certain viruses to bypass STB defenses. Recently described placental-villi associated maternal macrophages (PAMMs) have an unclear role in anti-pathogen defense; however, *ex vivo* studies suggest that they are microbicidal, can phagocytose bacteria, and do release pro-inflammatory cytokines following TLR stimulation. Finally, fetal macrophages *i.e.*, Hofbauer cells (HBCs) also have the capacity to secrete pro-inflammatory cytokines following TLR stimulation, can phagocytose pathogens *ex vivo* and may also use macrophage extracellular nets (METs) to kill bacteria. Whether PAMMs or HBCs employ Fc receptor mediated effector functions has not been explored but represents an intriguing mechanism of anti-pathogen defense given the high concentrations of maternal IgG in the intervillous and intravillous space. Figure was generated using Biorender.